

* Date of Revision: June 2023
Ca-antagonist

HERBESSER[®] TABLET 30MG

HERBESSER[®] 60 TABLET

(Diltiazem hydrochloride)



CONTRAINDICATIONS (HERBESSER is contraindicated in the following patients.)

- (1) Patients with severe congestive cardiac failure [Symptoms of cardiac failure may be aggravated].
- (2) Patients with second or third degree atrioventricular block or sick sinus syndrome (persistent sinus bradycardia) (less than 50 beats/ minute), sinus arrest, sinoatrial block, etc.) [Depression of cardiac stimulation and cardiac conduction may occur excessively].
- (3) Patients with a history of hypersensitivity to any of the ingredients of this product.
- (4) Pregnant women and women who may possibly be pregnant [See "Use during Pregnancy, Delivery or Lactation" section].
- (5) Patients receiving asunaprevir-containing product, ivabradine hydrochloride, or lomitapide mesylate [See "Drug Interactions" section].

DESCRIPTION

Product Name	HERBESSER TABLET 30MG	HERBESSER 60 TABLET
Active Ingredient (per tablet)	Diltiazem hydrochloride	
	30 mg	60 mg
Description	White tablets with mark TA 120 on one side and blank on other side	White tablets with mark TA 125 on one side and mark 60 on other side
Diameter	8.0 mm	8.0 mm
Thickness	3.45 mm	3.45 mm
Weight	0.19 g	0.185 g
Tablet Code	TA 120	TA 125

INDICATIONS

- Angina pectoris, variant angina pectoris
- Essential hypertension (mild to moderate)

DOSAGE AND ADMINISTRATION

- Angina pectoris, variant angina pectoris
The usual adult dosage for oral use is 30 mg of diltiazem hydrochloride three times a day (90 mg/day). The dosage may be increased to 60 mg three times a day (180 mg/day), if necessary.
- Essential hypertension (mild to moderate)
The usual adult dosage for oral used is 30 to 60 mg of diltiazem hydrochloride three times a day (90 – 180 mg/day). The dosage may be adjusted depending on the patient's age and symptoms.

PRECAUTIONS

1. Careful Administration (HERBESSER should be administered with care in the following patients.)

- 1) Patients with congestive cardiac failure. [Symptoms of cardiac failure may be aggravated.]
- 2) Patients with severe bradycardia (less than 50 beats/min) or first grade atrioventricular block. [Depression of cardiac stimulation and cardiac conduction may occur excessively.]
- 3) Patients with excessively low blood pressure [Blood pressure may be further decreased.]
- 4) Patients with severe hepatic or renal impairment. [The metabolism and excretion of this product may be prolonged, and effects may be intensified.]

2. Important Precautions

- 1) It has been reported that abrupt cessation of calcium antagonists may aggravate the patient's symptoms. In case where cessation of this product is required, the dosage should be reduced gradually under careful observation of the patient. Patients should be instructed not to discontinue taking this product without consulting physicians.
- 2) Since dizziness, etc. may occur due to antihypertensive action of this product, patients should be cautioned against engaging in potentially hazardous activities, such as working at altitude or driving motor vehicles.
- 3) Prolonged QT and ventricular arrhythmia have been reported in coadministration of terfenadine with other antiarrhythmic agents (disopyramide phosphate).

3. Drug Interactions

This product is metabolized mainly by cytochrome P450 3A4 (CYP3A4) metabolizing enzyme.

(1) Contraindications for Co-administration (Do not co-administer with the following.)

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Asunaprevir Daclatasvir hydrochloride/ asunaprevir / beclabuvir hydrochloride	The blood concentration of asunaprevir increased. Hepatobiliary disorder may occur and become severe.	This product inhibits CYP3A, and the metabolism of these drugs is inhibited.
Ivabradine hydrochloride	Excessive bradycardia may occur.	This product inhibits CYP3A, the metabolism of ivabradine is inhibited, and the blood concentration of ivabradine is increased. The heart rate reducing effect of ivabradine hydrochloride is potentiated additively.
Lomitapide mesylate	The blood concentration of lomitapide mesylate may be markedly increased.	This product inhibits CYP3A, and the metabolism of lomitapide is inhibited.

(2) Precautions for coadministration (HERBESSER should be administered with care when co-administered with the following drugs.)

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Drugs with antihypertensive effects (antihypertensive drugs, nitric acid preparation, etc.)	Antihypertensive effects may be intensified. Blood pressure should be measured periodically to adjust the dosage.	Antihypertensive effects may be intensified additively.
Beta blockers (bisoprolol fumarate, propranolol hydrochloride, atenolol, etc.)	Bradycardia, atrioventricular block, sinoatrial block, etc. may occur. Pulse rate should be measured periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Depression of cardiac stimulation and cardiac conduction, negative inotropic effects, and antihypertensive effects may be intensified additively. Particular attention should be given to triple therapy using this product with digitalis preparation and beta blocker or rauwolfia preparation.
Rauwolfia preparations (reserpine, etc.)		
Digitalis preparations (digoxin, methyl digoxin)	Bradycardia, atrioventricular block, etc. may occur. In addition, toxic symptoms (nausea, vomiting, headache, dizziness, abnormal vision, etc.) including above arrhythmic symptoms may occur due to an increased blood concentration of digitalis preparation. Presence or absence of digitalis toxicity should be observed periodically, and electrocardiogram should be performed. In addition, blood concentration of digitalis preparation should be measured as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Depression of cardiac stimulation and cardiac conduction may be intensified additively. Particular attention should be given to triple therapy using this product with digitalis preparation and beta blocker. This product may increase blood concentrations of digitalis preparations.
Antiarrhythmic agents (amiodarone hydrochloride, mexiletine hydrochloride, etc.)	Bradycardia, atrioventricular block, sinus arrest, etc. may occur. Pulse rate should be measured periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Depression of cardiac stimulation and cardiac conduction may be intensified additively.
Fingolimod hydrochloride	Severe bradycardia or heart block may occur by concomitant use of this product during the initiation of fingolimod by hydrochloride.	Both diltiazem hydrochloride and fingolimod hydrochloride may induce bradycardia or heart block.

Aprindine hydrochloride	Symptoms (bradycardia, atrioventricular block, sinus arrest, tremor, dizziness, light-headedness, etc.) may occur due to increased blood concentrations of both drugs. Clinical symptoms should be observed periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Each drug may affect a common metabolizing enzyme (cytochrome P450), and increase blood concentration of each drugs.
Dihydropyridine calcium antagonists (nifedipine, amlodipine besilate, etc.)	Symptoms (intensified antihypertensive effects, etc.) may occur due to increased blood concentration of dihydropyridine calcium antagonist. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	This product may inhibit the metabolizing enzyme (cytochrome P450) of these drugs, and increase their blood concentrations.
Simvastatin	Rhabdomyolysis or myopathy may occur due to increased blood concentration of simvastatin. Clinical symptoms should be observed periodically. If any abnormalities are observed, administration should be discontinued.	
Triazolam	Symptoms (prolonged sleeping time, etc.) may occur due to increased blood concentration of triazolam. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Midazolam	Symptoms (intensified sedative and hypnotic effect, etc.) may occur due to increased blood concentration of midazolam. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Carbamazepine	Symptoms (sleepiness, nausea, vomiting, dizziness, etc.) may occur due to increased blood	

	concentration of carbamazepine, Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Selegiline hydrochloride	Effects and toxicity of selegiline hydrochloride may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Theophylline	Symptoms (nausea, vomiting, headache, insomnia, etc.) may occur due to increased blood concentration of theophylline. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Cilostazol	Effects of cilostazol may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Apixaban	Effects of apixaban may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Vinorelbine tartrate	Effects of vinorelbine tartrate may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Cyclosporin	Symptoms (renal disorder, etc.) may occur due to increased blood concentration of cyclosporin. Clinical symptoms should be observed periodically, and blood concentration of cyclosporin should be measured. If any abnormalities are observed, the	

	dosage should be reduced or administration should be discontinued	
Tacrolimus hydrate	Symptoms (renal disorder, etc.) may occur due to increased blood concentration of tacrolimus. Clinical symptoms should be observed periodically, and blood concentration of tacrolimus should be measured. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Phenytoin	Symptoms (ataxia, dizziness, nystagmus, etc.) may occur due to increased blood concentration of phenytoin. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued. Effect of this product may be attenuated.	This product may inhibit the metabolizing enzyme (cytochrome P450) of phenytoin. and increase blood concentrations of phenytoin. In addition: phenytoin may stimulate metabolism of this product, and decrease blood concentration of this product
Cimetidine	Symptoms (intensified antihypertensive effect, bradycardia, etc.) may occur due to increased blood concentration of this product. Clinical symptoms should be observed periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	These drugs may inhibit the metabolizing enzyme (cytochrome P450) of this product, and increase blood concentration of this product.
HIV protease inhibitors (ritonavir, saquinavir mesylate, etc.)		
Rifampicin	Effects of this product may be attenuated. Clinical symptoms should be observed periodically, and if possible, blood concentration of this product should be measured. If any abnormalities are observed, appropriate therapeutic measures such as changing to other drugs or increasing the dosage of this product should be taken.	Rifampicin may induce the metabolizing enzyme (cytochrome P450) of this product, and decrease blood concentrations of this product.
Anesthetics drugs (isoflurane, enflurane, halothane, etc.)	Bradycardia, atrioventricular block, sinus arrest, etc. may occur. Electrocardiogram should be monitored. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Depression of cardiac stimulation and cardiac conduction may be intensified additively.

Muscle relaxants (pancuronium bromide, vecuronium bromide, etc.)	Effects of muscle relaxants may be intensified. Caution should be exercised to muscle relaxants action. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	This product may inhibit the acetylcholine release from the presynaptic terminals at the neuromuscular junction.
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4. Adverse Reactions

Adverse reactions were reported in 442 (4.6 %) of 9,630 patients. The major adverse reactions were gastrointestinal symptoms in 1.4% (stomach discomfort in 0.2%, constipation in 0.2%, abdominal pain in 0.1%, etc.), cardiovascular symptoms in 1.4% (dizziness in 0.5%, bradycardia in 0.4%, facial flushing in 0.2%, atrioventricular block in 0.2%, etc.), hypersensitivity in 1.2%, headache in 0.2%, etc. (Data collected from the time of approval up to December 1990).

(1) Clinically significant adverse reactions (rarely: <0.1%, no adverb: incidence unknown because of spontaneous reports)

- 1) **Complete atrioventricular block, severe bradycardia** (early symptom: bradycardia, dizziness, light-headedness, etc.) may occur rarely. If any abnormalities are observed, administration should be discontinued, and atropine sulfate hydrate, isoprenaline, etc. should be administered or appropriate therapeutic measures such as cardiac pacing should be taken as needed.
- 2) **Congestive cardiac failure** may occur. If any abnormalities are observed, administration should be discontinued, and appropriate therapeutic measures such as administration of cardiotonic drug, etc. should be taken.
- 3) **Oculomucocutaneous syndrome (Steven-Johnson syndrome), toxic epidermal necrolysis (Lyell syndrome), erythroderma (exfoliative dermatitis), acute generalized exanthematous pustulosis** may occur. If erythema, blister, pustule, pruritis, fever, enanthema, etc. occur, administration should be discontinued, and appropriate therapeutic measures should be taken.
- 4) **Hepatic function disorder or jaundice** with increased AST (GOT), ALT (GPT) or γ -GTP may occur. The patient's conditions should be observed carefully. If any abnormalities are observed, administration should be discontinued, and appropriate therapeutic measures should be taken.

(2) Other adverse reactions

If any adverse reactions are observed, appropriate therapeutic measures such as discontinuation of treatment should be taken.

Incidence Type	5% > $\geq$ 0.1%	< 0.1%	Incidence unknown
Cardiovascular	Bradycardia, atrioventricular block, facial flushing, dizziness	Sinus arrest, decreased blood pressure, palpitation, chest pain, oedema	Sinoatrial block
Psychoneurotic	Malaise, headache, headache dull	Cramps in the calves, feelings of weakness, sleepiness, insomnia	Parkinsonian-like symptom

Hepatic	Increased AST (GOT), increased ALT(GPT)	Jaundice	Increased Al-P, increased LDH, increased γ -GTP, hepatic hypertrophy
Hypersensitivity	Rash	Pruritis, multiforme erythematous, rash, urticaria	Photosensitivity, pustule
Gastrointestinal	Stomach discomfort, constipation, abdominal pain, heartburn, anorexia, nausea	Soft stool, diarrhea, thirst.	-----
Hematologic	-----	-----	Decreased platelets count, decreased white blood cell count
Other	-----	-----	Gingival hypertrophy, gynaecomastia: numbness

5. Use in the Elderly

Excessive decrease in blood pressure is generally considered undesirable in elderly patients. Therefore, HERBESSER should be administered with care such as starting from a lower dosage while, carefully monitoring the patient's condition.

6. Use during Pregnancy, Delivery or Lactation

- 1) HERBESSER should not be administered to pregnant women or women who may possibly be pregnant. [Animal studies have shown teratogenicity (mice: skeletal abnormality, appearance abnormality) and embryotoxicity (mice, rats: fatality).]
- 2) Use of this product in lactating women is not recommended. If treatment with this product is judged to be essential, breast feeding must be discontinued during treatment. [It has been reported that diltiazem is excreted in human breast milk.]

7. Pediatric Use

The safety of HERBESSER in children has not been established.

8. Overdosage

Symptoms:

Bradycardia, complete atrioventricular block, cardiac failure, hypotension, etc. may occur after overdosage of HERBESSER. These symptoms have been also reported as adverse reactions.

Treatment:

In case of overdosage, administration should be discontinued, and gastric lavage should be performed to remove the product as needed, and the following appropriate therapeutic measures should be taken.

- 1) Bradycardia, complete atrioventricular block
Atropine sulfate hydrate, isoprenaline, etc. should be administered or cardiac pacing should be taken.
- 2) Cardiac failure, hypotension
Cardiotonic drug, vasopressor, infusion, etc. should be administered or assisted circulation should be performed.

9. Precautions concerning Use

1) Precautions regarding dispensing:

For drugs that are dispensed in a press-through package (PTP), instruct the patients to remove the drug from the PTP sheet prior to use. [It was reported that, if the PTP sheet is swallowed, the sharp corners of the sheet may puncture the esophageal mucosa, resulting in severe complications such as mediastinitis.]

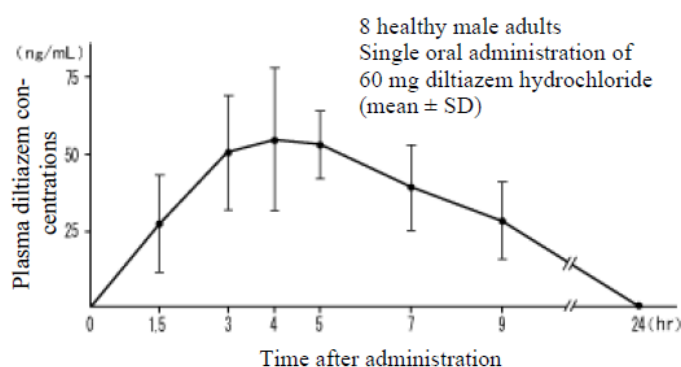
2) Precautions during oral administration:

This product should be taken without chewing. [It may impair the sustained-release characteristics of this product.]

PHARMACOKINETICS

1. Plasma concentration

Plasma concentration of this product after oral administration of two tablets of HERBESSER Tablets 30mg (60mg of diltiazem hydrochloride) to healthy male adults reached maximum at 3 to 5 hours after the administration, and thereafter decreased with an elimination half-life about 4.5 hours. In repeated oral administration, the plasma concentration reached a steady state on the 2nd day or later. The plasma concentration was about 40 ng/mL at 2 to 4 hours after the administration in patients treated with long-term administration of 90 mg/day in three divided doses.¹⁾



2. Metabolism

When this product was administered orally to healthy male adults, the main metabolic pathways were oxidative deaminization, oxidative demethylation, deacetylation, and conjugation.²⁾

CLINICAL STUDIES

1. Angina pectoris, variant angina

The usefulness of this product was proven for angina pectoris in a double-blind comparative study, single-blind comparative study, and open clinical study³⁻⁶⁾ as well as for variant angina in an open clinical study including an evaluation of Holter electrocardiogram⁷⁾.

2. Essential hypertension

The usefulness of this product was proven for essential hypertension in four double-blind comparative studies using placebo, reserpine, and propranolol as controls⁸⁻¹¹⁾.

NON-CLINICAL STUDIES

Toxicity

1) Single dose toxicity (LD₅₀ mg/kg)¹²⁾

	p.o.		s.c.		i.v.	
	male	female	male	female	male	female
ddY-strain mice	740	640	260	280	61	58
Wistar-strain rats	560	610	520	550	38	39

2) Repeated does toxicity

When 2, 10, 25 and 125 mg/kg/day, and 10, 20, and 40 mg/kg/day of diltiazem hydrochloride were orally given respectively to SD-strain rats and beagle dogs for six successive months, the mortal cases in the 125 mg/kg/day group, and hepatic and renal impediments in the 25 and 125 mg/kg/day group were observed in rats, but no influences were observed in the 2 and 10 mg/kg/day group. The mortal cases and abnormalities in electrocardiograms in the 40 mg/kg group were observed in beagle dogs and the transient increase in GOT and GPT in the 20 mg/kg/day group was observed¹³⁻¹⁴⁾.

3) Teratogenicity

1. Administration before mating, and during pregnancy and lactation:
12.5, 25, 50, and 100 mg/kg/day of diltiazem hydrochloride were orally given to CFY strain rats. Adverse effects on reproductive function in parents, and lethality, teratogenicity, and growth retardation in fetuses and pups were not observed¹⁵⁾
2. Administration during the period of fetal organogenesis 10, 25, 50, 100, 200, and 400 mg/kg/day, and 10, 50, 100, 200 and 400 mg/kg/day of diltiazem hydrochloride were orally given respectively to 1CR-JCL strain mice and Wistar strain rats. The fetocidal effects was observed in the 50, 100, 200, 400 mg/kg/day groups of mice, but it was not observed even in the 400 mg/kg/day group of rats¹⁷⁾
3. Administration during the peri-and post-natal periods: 10, 50, 100, 200, and 400 mg/kg/day of diltiazem hydrochloride were orally given to Wistar strain rats. As general conditions in dams changed for the worse, decreases in the viability rate and body weight gain of pups were observed in the 200 and 400 mg/kg/day group, but no remarkable change was observed in the 10, 50, and 100 mg/kg/day groups⁵⁾

4) Antigenicity

The antigenicity of diltiazem hydrochloride was not observed in guinea pigs, mice, and rats¹⁷⁾

5) Mutagenicity

The mutagenicity of diltiazem hydrochloride was not observed either by the repair and reserve mutation test with bacteria, the chromosomal aberration test with cultured cells of mammalia, or the micronucleus test with mice¹⁸⁻²⁰⁾

6) Carcinogenicity

The carcinogenicity of diltiazem hydrochloride was not observed by the test with mice²¹⁾

PHARMACOLOGY

Diltiazem hydrochloride dilates blood vessels, improves myocardial ischemia, and exerts antihypertensive effect by inhibiting calcium channel influx to cells in vascular smooth muscles such as coronary vessels, peripheral vessels, etc.

1. Action on myocardial ischemia

1) Improvement of myocardial oxygen demand and supply balance

- (1) It dilates large coronary vessels and collateral channels, and increases blood flow to myocardial ischaemic lesion (dogs).²²⁻²⁵⁾
- (2) It inhibits coronary artery spasm (monkeys and humans).^{26, 27)}
- (3) It decreases myocardial oxygen consumption by afterload reduction due to peripheral vasodilation and decreased heart rate without decreasing cardiac output (dogs).²⁸⁾

2) Myocardial protective action

It maintains cardiac function and myocardial energy metabolism, and reduces the extension of infarct lesions by inhibiting calcium channel influx to cells at the time of myocardial ischaemia (rats).²⁹⁾

2. Action on blood pressure

- 1) It has almost no effect on normal blood pressure, while decreases high blood pressure gradually (rats and humans)³⁰⁻³²⁾, and inhibits exercise-induced blood pressure elevation (humans).³³⁾
- 2) It does not decrease cerebral and renal blood flow, while decreases blood pressure (dogs and humans).³⁴⁻³⁷⁾
- 3) It inhibits myocardial and vascular hypertrophies along with decreased blood pressure (rats).³⁸⁾

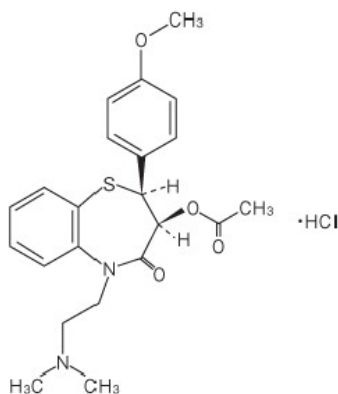
3. Effects on cardiac stimulation and cardiac conduction

It slightly prolongs sinus node spontaneous cycle length and slows atrioventricular nodal conduction (AH interval), while it does not effect His-Purkinje system conduction (HV interval) (dogs and humans).^{28,39,40)}

PHYSICOCHEMISTRY

Nonproprietary name: Diltiazem Hydrochloride (JAN)

Chemical name: (2S,3S)-5-[2-(Dimethylamino) ethyl]-2-(4-methoxyphenyl)-4-oxo-2,3,4,5-tetrahydro-1,5-benzothiazepin-3-yl-acetate monohydrochloride



C₂₂H₂₆N₂O₄S · HCl : 450.98

Description:

- Diltiazem hydrochloride occurs as white crystals or crystalline powder. It is odorless
- It is very soluble in formic acid, freely soluble in water, in methanol and in chloroform, sparingly soluble in acetonitrile, slightly soluble in acetic anhydride and in ethanol (99.5), and practically insoluble in diethyl ether.
- Optical rotation $[\alpha]_D^{20}$: +115 - +120° (after drying, 0.20 g, water, 20 mL, 100 mm).
- Melting point: 210 - 215°C (with decomposition)

STORAGE AND HANDLING

- 1) Caution: Use only pursuant to prescription or direction of a physician
- 2) Caution for administration: Swallow the tablets without chewing.
- 3) Storage: Do not store above 30°C. Avoid exposure to direct sunlight.
- 4) Expiration date: Indicated on the package

PACKAGING

HERESSER Tablet 30mg:

Boxes of 30 tablets (10 tablets x 3) in PTP

Boxes of 100 tablets (10 tablets x 10) in PTP

Boxes of 1,000 tablets (10 tablets x 100) in PTP

Boxes of 600 tablets (20 boxes of 30) in PTP

HERBESSER 60 Tablet:

Boxes of 100 tablets (10 tablets x 10) in PTP

Boxes of 1,000 tablets (10 tablets x 100) in PTP

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